Tightness of the rescaled small-parameter LQG geodesic

Abstract

Let P_γ be the γ-Liouville quantum gravity geodesic from 0 to 1, parametrized at constant LQG-metric speed, and set

$$X_\gamma(t) := \frac{P_\gamma(t) - t}{\gamma^{2/3}}, \qquad t \in [0,1].$$

We prove that $(X_\gamma)_{\gamma \downarrow 0}$ is tight in $C([0,1], \mathbb{R}^2)$. The scalar tightness of the maximal deviation $M_\gamma/\gamma^{2/3}$ is not by itself sufficient. The proof uses one scalar estimate from the earlier deviation argument, namely $\mathbb{E}M_\gamma \lesssim \xi^{2/3}$, together with two intermediate ingredients from that argument: the good-event clock geometry and the frequency-localized single-probe estimate. These estimates give tight control of the low-frequency part of the transverse displacement, while the unresolved high-frequency part is $o_{\mathbb{P}}(\xi^{2/3})$ in the uniform norm.

1 Statement and parametrization

Let h be a whole-plane GFF with the same normalization as in the earlier two-thirds deviation argument. Let D_h^γ be the γ-LQG metric and let P_γ be the almost surely unique D_h^γ-geodesic from 0 to 1. We use the constant-speed parametrization

$$D_h^\gamma(0, P_\gamma(t)) = t D_h^\gamma(0,1), \qquad 0 \le t \le 1. \tag{1}$$

This convention is essential: the assertion below is false for arbitrary parametrizations of the same unparametrized curve.

Write

$$P_\gamma(t) = x_\gamma(t) + i y_\gamma(t), \qquad M_\gamma := \max_{t \in [0,1]} \operatorname{dist}(P_\gamma(t), [0,1]).$$

As before, let

$$\xi = \frac{\gamma}{d_\gamma}, \qquad \delta = \xi^{2/3}, \qquad L = \log(1/\xi).$$

The small-γ estimate $d_\gamma = 2 + O(\gamma)$ gives

$$\xi \sim \frac{\gamma}{2}. \tag{2}$$

Theorem 1.1. *The family*

$$X_\gamma(t) = \frac{P_\gamma(t) - t}{\gamma^{2/3}}, \qquad t \in [0,1],$$

is tight in $C([0,1], \mathbb{R}^2)$ as $\gamma \to 0$.

2 Inputs from the two-thirds deviation proof

We record precisely what is imported from the earlier proof. In particular, we do not use the previously proved upper-tail estimate for M_γ/δ.

Let $\widehat{D}_h$ denote the deterministic point-distance normalization used there and put

$$\Delta_3 := \sup_{u,v\in B_3(0)} \left|\widehat{D}_h(u,v) - |u-v|\right|.$$

Multiplying the metric by a positive scalar does not change its geodesics or the parametrization (1) after normalization to the time interval $[0,1]$.

The following four facts were established in the earlier argument.

Proposition 2.1 (Imported estimates)**.** *There are constants $C<\infty$, $c_\beta>0$, and $\gamma_0>0$ with the following properties for $0<\gamma<\gamma_0$.*

First, with

$$m_\gamma := \mathbb{E}M_\gamma,$$

one has the scalar mean bound

$$m_\gamma \le C\delta. \tag{3}$$

Second, there is an event $\mathcal{G}_\gamma$ such that

$$\mathbb{P}(\mathcal{G}_\gamma^c) \le C\xi^{20}, \qquad \Delta_3 \le C\xi L \quad \text{on } \mathcal{G}_\gamma. \tag{4}$$

On $\mathcal{G}_\gamma$, the constant-speed parametrization satisfies

$$|x_\gamma(t)-t| \le C\Delta_3, \qquad |y_\gamma(t)-y_\gamma(s)| \le C\sqrt{|t-s|\Delta_3} + C\Delta_3. \tag{5}$$

Third, let $\ell_j = 2^{-j}$. Fix the smooth cutoff χ and band multiplier

$$\rho(s) = \chi(s) - \chi(2s)$$

used in the earlier proof. If

$$b_{\gamma,k} := 2\int_0^1 y_\gamma(t)\sin(k\pi t)\,dt,$$

define

$$B_{\gamma,j}(a) := \sum_{k\ge 1} \rho(k\ell_j) b_{\gamma,k}\sin(k\pi a). \tag{6}$$

These are trigonometric polynomials of degree at most $2\ell_j^{-1}$. There are deterministic sets $\mathcal{A}_j \subset [0,1]$ such that

$$|\mathcal{A}_j| \le C\ell_j^{-1}, \qquad \|B_{\gamma,j}\|_\infty \le 2\max_{a\in\mathcal{A}_j}|B_{\gamma,j}(a)|. \tag{7}$$

Moreover, for every J,

$$y_\gamma = \sum_{j=0}^{J} B_{\gamma,j} + R_{\gamma,J}, \qquad \|R_{\gamma,J}\|_\infty \le C\omega_{y_\gamma}(C\ell_J), \tag{8}$$

where

$$\omega_f(r) := \sup_{|t-s|\le r} |f(t)-f(s)|.$$

Indeed, the identity $\sum_{j=0}^{J}\rho(k\ell_j)=\chi(k\ell_J)$ *makes the sum in* (8) *the low-pass sine-series cutoff at frequency* ℓ_J^{-1}*, and the cutoff kernel estimate from the earlier proof gives the stated remainder bound. Since* $y_\gamma(0)=y_\gamma(1)=0$*, one also has* $B_{\gamma,j}(0)=B_{\gamma,j}(1)=0$*.*

Fourth, take

$$\ell_{J(\gamma)}\asymp\xi^{3/8} \tag{9}$$

and set $\beta=1/8$*. Uniformly for*

$$\delta\le R\le\xi^{1/2}L, \tag{10}$$

the frequency-localized probe estimate gives

$$\begin{aligned}&\mathbb{P}\left[\exists j\le J(\gamma),\ a\in\mathcal{A}_j:|B_{\gamma,j}(a)|>c_\beta R\ell_j^\beta,\ \mathcal{G}_\gamma\right]\\&\qquad\le C\left(\frac{\xi^2}{R^3}+\frac{\xi^2(m_\gamma+\xi L^2)}{R^4}+\frac{\xi^{49/32}L^2}{R^2}\right).\end{aligned} \tag{11}$$

Remark 2.2. *The only scalar estimate about* M_γ *used below is* (3)*. In particular, the previously proved bound*

$$\limsup_{\gamma\to0}\mathbb{P}[M_\gamma>A\delta]\lesssim A^{-3}+A^{-4}$$

is not used. What is additionally needed is the stronger frequency-localized information (11)*, which was an intermediate step in the proof of that scalar tail bound.*

3 The horizontal component is negligible

Set

$$Z_\gamma(t):=\frac{P_\gamma(t)-t}{\delta}.$$

On $\mathcal{G}_\gamma$, (4) and (5) imply

$$\sup_{t\in[0,1]}\left|\frac{x_\gamma(t)-t}{\delta}\right|\le C\frac{\xi L}{\xi^{2/3}}=C\xi^{1/3}L.$$

Since $\mathbb{P}(\mathcal{G}_\gamma^c)\to0$,

$$\|(x_\gamma-\mathrm{id})/\delta\|_\infty\xrightarrow[\gamma\to0]{\mathbb{P}}0. \tag{12}$$

Thus it remains to prove tightness of y_γ/δ in $C([0,1],\mathbb{R})$.

4 High frequencies are negligible

Let $J=J(\gamma)$ be as in (9), and write

$$S_\gamma:=\sum_{j=0}^{J}B_{\gamma,j},\qquad R_\gamma:=R_{\gamma,J}.$$

On $\mathcal{G}_\gamma$, (5) and (9) give

$$\begin{aligned}\|R_\gamma\|_\infty &\le C\omega_{y_\gamma}(C\ell_J)\\ &\le C\sqrt{\ell_J\Delta_3}+C\Delta_3\\ &\le C\xi^{11/16}L^{1/2}+C\xi L.\end{aligned}$$

Consequently,

$$\frac{\|R_\gamma\|_\infty}{\delta}\le C\xi^{1/48}L^{1/2}+C\xi^{1/3}L\longrightarrow 0\qquad\text{on }\mathcal{G}_\gamma. \tag{13}$$

Together with $\mathbb{P}(\mathcal{G}_\gamma^c)\to 0$, this yields

$$\|R_\gamma/\delta\|_\infty \xrightarrow[\gamma\to 0]{\mathbb{P}} 0. \tag{14}$$

It therefore suffices to prove tightness of S_γ/δ.

5 Tightness of the low-frequency part

Fix $K\ge 1$. Define $\mathcal{E}_{\gamma,K}$ to be the event

$$|B_{\gamma,j}(a)|\le c_\beta K\delta\ell_j^\beta\quad\text{for every } j\le J(\gamma),\ a\in\mathcal{A}_j. \tag{15}$$

For each fixed K, the choice $R=K\delta$ lies in the range (10) for all sufficiently small γ, since

$$\frac{K\delta}{\xi^{1/2}L}=\frac{K\xi^{1/6}}{L}\longrightarrow 0.$$

Hence (11) and (3) give

$$\begin{aligned}\mathbb{P}(\mathcal{E}_{\gamma,K}^c)&\le\mathbb{P}(\mathcal{G}_\gamma^c)+C\left(K^{-3}+\frac{\xi^2(m_\gamma+\xi L^2)}{K^4\delta^4}+\frac{\xi^{49/32}L^2}{K^2\delta^2}\right)\\ &\le C\xi^{20}+C\left(K^{-3}+K^{-4}+K^{-4}\xi^{1/3}L^2+K^{-2}\xi^{19/96}L^2\right).\end{aligned} \tag{16}$$

Therefore

$$\limsup_{\gamma\to 0}\mathbb{P}(\mathcal{E}_{\gamma,K}^c)\le C(K^{-3}+K^{-4}). \tag{17}$$

On $\mathcal{E}_{\gamma,K}$, (7) gives

$$\|B_{\gamma,j}\|_\infty\le CK\delta\ell_j^\beta. \tag{18}$$

Since $B_{\gamma,j}$ has degree at most $2\ell_j^{-1}$, Bernstein's inequality gives

$$\left\|B'_{\gamma,j}\right\|_\infty\le CK\delta\ell_j^{\beta-1}. \tag{19}$$

It follows that, for $s,t\in[0,1]$ and $r=|t-s|$,

$$\frac{|B_{\gamma,j}(t)-B_{\gamma,j}(s)|}{\delta}\le CK\ell_j^\beta\min\left\{1,\frac{r}{\ell_j}\right\}. \tag{20}$$

We use the elementary dyadic estimate

$$\sum_{j\geq 0} \ell_j^{\beta} \min\left\{1, \frac{r}{\ell_j}\right\} \leq C_\beta r^\beta, \qquad 0 \leq r \leq 1. \tag{21}$$

Indeed, split the sum at the unique dyadic scale comparable to r. On scales $\ell_j \geq r$, the sum is bounded by

$$r \sum_{\ell_j \geq r} \ell_j^{\beta-1} \leq C_\beta r^\beta,$$

and on scales $\ell_j < r$ it is bounded by

$$\sum_{\ell_j < r} \ell_j^{\beta} \leq C_\beta r^\beta.$$

Combining (20) and (21), we obtain on $\mathcal{E}_{\gamma,K}$

$$\left|\frac{S_\gamma(t) - S_\gamma(s)}{\delta}\right| \leq CK|t-s|^{1/8}. \tag{22}$$

Since every $B_{\gamma,j}(0) = 0$, also $S_\gamma(0) = 0$. Thus

$$\|S_\gamma/\delta\|_{C^{1/8}} \leq CK \quad \text{on } \mathcal{E}_{\gamma,K}, \tag{23}$$

where any standard norm equivalent to

$$\|f\|_\infty + \sup_{s\neq t} \frac{|f(t)-f(s)|}{|t-s|^{1/8}}$$

may be used.

For $K \geq 1$, let

$$\mathcal{K}_K := \{f \in C([0,1],\mathbb{R}) : f(0) = 0,\ \|f\|_{C^{1/8}} \leq CK\}.$$

By Arzelà–Ascoli, $\mathcal{K}_K$ is compact in $C([0,1],\mathbb{R})$. Equations (17) and (23) imply

$$\limsup_{\gamma\to 0} \mathbb{P}[S_\gamma/\delta \notin \mathcal{K}_K] \leq C(K^{-3} + K^{-4}).$$

Given $\varepsilon > 0$, choose K so large that $C(K^{-3}+K^{-4}) < \varepsilon/2$, and then choose γ sufficiently small that the remaining terms in (16) are less than $\varepsilon/2$. Hence

$$(S_\gamma/\delta)_{\gamma\downarrow 0} \quad \text{is tight in } C([0,1],\mathbb{R}). \tag{24}$$

Remark 5.1. *Equation* (23) *is a H"older estimate only for the low-frequency truncation S_γ/δ. The remainder R_γ/δ is shown to vanish in the uniform norm, but no uniform H"older estimate for it is asserted. Accordingly, the conclusion below is tightness in $C([0,1])$, not tightness in $C^{1/8}([0,1])$.*

6 Conclusion

We use the standard fact that if U_γ is tight in a metric space and $d(U_\gamma, V_\gamma) \to 0$ in probability, then V_γ is tight. From (14) and (24),

$$\frac{y_\gamma}{\delta} = \frac{S_\gamma}{\delta} + \frac{R_\gamma}{\delta}$$

is tight in $C([0,1], \mathbb{R})$. Combining this with (12) shows that

$$\left(\frac{P_\gamma - \mathrm{id}}{\xi^{2/3}}\right)_{\gamma \downarrow 0} \quad \text{is tight in } C([0,1], \mathbb{R}^2). \tag{25}$$

Finally,

$$\frac{P_\gamma - \mathrm{id}}{\gamma^{2/3}} = d_\gamma^{-2/3} \frac{P_\gamma - \mathrm{id}}{\xi^{2/3}},$$

and $d_\gamma^{-2/3} \to 2^{-2/3}$ by (2). Multiplication by this deterministic convergent scalar preserves tightness. This proves Theorem 1.1.

Remark 6.1. *The proof identifies exactly why scalar control of M_γ is insufficient by itself. The mean estimate* (3) *closes the second term in the probe bound, but equicontinuity comes from the frequency-localized estimate* (11)*. The low-frequency bands are uniformly Hölder on events of arbitrarily high probability, while the remaining high frequencies have vanishing amplitude in the uniform norm.*